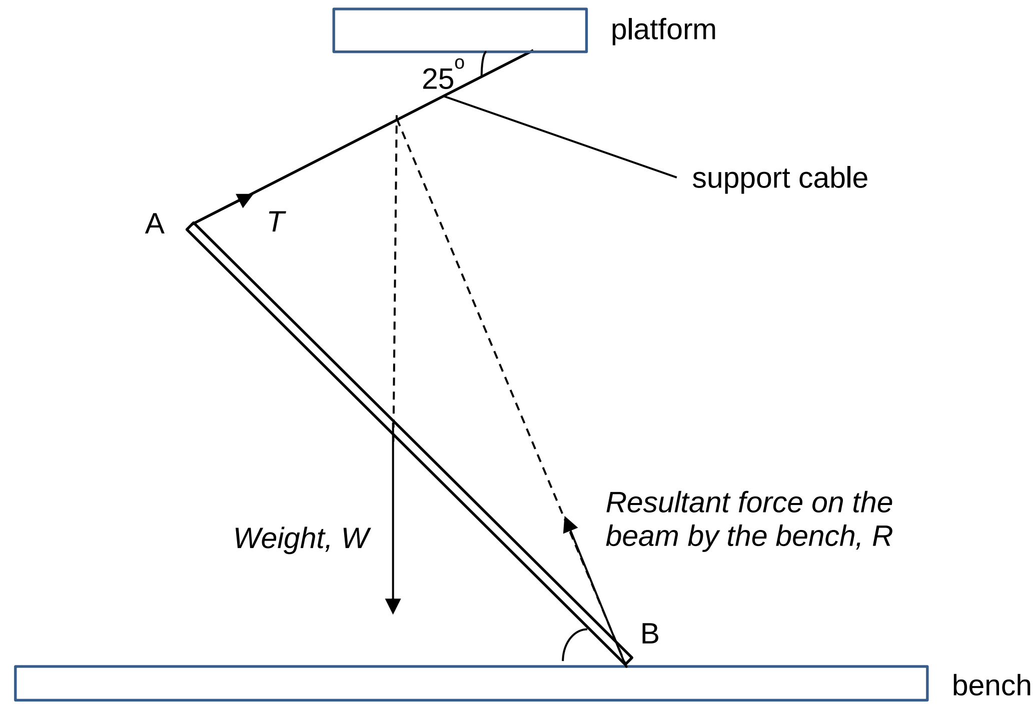
8	(a)	The **net external force acting on a body is zero.** The **net torque on the body about any point is zero**.		B1 B1
	(b)	(i)	 Correct labelling of the 3 forces and point of contact – 1 mark Correct direction (check 3 forces intersect at a common point)	M1 A1
		(ii)	Taking moments about B, and by the principle of moments Sum of clockwise moments = Sum of anti-clockwise moments $T\sin 67^o \times L = mg \times (\frac{L}{2}\cos 42^o)$ $T = \frac{mg\cos 42^o}{2\sin 67^o} = \frac{1.5(9.81)\cos 42^o}{2\sin 67^o} = 5.939\,\text{N} = 5.9\,\text{N}$	B1 M1 A0
		(iii)	$\sum F_x = 0 \Rightarrow R_x = T_x = 5.939\cos 25^o = 5.383\,\text{N}$ $\sum F_Y = 0 \Rightarrow R_Y + T_Y = W$ $R_Y = W - T_Y = 1.5(9.81) - 5.939\sin 25^o = 12.21\,\text{N}$ $\therefore R = \sqrt{(R_X)^2 + (R_Y)^2} = \sqrt{5.383^2 + 12.21^2} = 13.3\,\text{N}$ $\tan\theta = \frac{F_Y}{F_X} = \frac{12.21}{5.383}$ $\theta = 66.2^o$ Direction = 66.2º clockwise above the horizontal. (need a short description)	M1 M1 A1 A1

	(c)	**(i)**	By Newton's 2nd law, $F_{net} = ma$ $mg\sin\theta = ma$ $a = g\sin\theta = 9.81\sin 42^o = 6.56\ \text{m s}^{-2}$	 M1 A1
		(ii)	Acceleration of the ball **remains the same** as it is **independent of mass or weight** as shown in the equation in (c)(i).	B1 B1
		(iii)	From start to top of circular track, Loss in GPE = Gain in KE $mg\left(0.58\sin 42^o - 0.34\right) = \frac{1}{2}mv_{top}^2 - 0$ $v_{top} = \sqrt{2\left(9.81\right)\left(0.58\sin 42^o - 0.34\right)}$ $= 0.9714 = 0.971\ m\,s^{-1}$ Consider the forces acting on the ball at the top of the circular track, $mg + N = \frac{mv^2}{r}$ To find the minimum speed at the top of the circular track, consider N→0, $v_{\min} = \sqrt{rg} = \sqrt{0.17\left(9.81\right)} = 1.29\ m\,s^{-1}$ Since $v_{top} < v_{\min}$, the ball is unable to make a complete loop around the circular track.	M1 A1 A1 A1
		(iv)	The ball will **remain in contact with the circular track at least clearing half of the height of the circular track or ¼ of the circular track** and then move in a **parabolic path when it leaves the tracks**.	B1 B1
			Total:	20

9	**(a)**	The magnetic flux linkage is defined as the **product of the number of turns** of the coil	B1